

NIELIT Virtual Academy

COURSE PROSPECTUS

Name of the Course: Foundation course in ML using Python

Course Code: DS32

Mode of Conduction: Online

Last date of registration: 24th April 2025

Starting Date: 28th April 2025

Duration: 90 hours

Objective of the Course

The objective of this course is to enabling the student with basic knowledge on the techniques to build an intellectual machine for making decisions behalf of humans. This course covers the techniques on how to make learning by a model, how it can be evaluated, what are all different algorithms to construct a learning model.

Prerequisites

- ✓ Candidate must have latest computer/laptop with preferably 4 GB RAM or higher and Graphics Card (2 GB)
- ✓ Internet connection with good speed (preferably 2Mbps or higher)

Outcome of the Course:

Implement and analyze existing learning algorithms, including methods for classification, regression, structured prediction, clustering, and representation learning.

- ✓ Integrate multiple aspects of practical implementation of machine learning: Data preprocessing, Regularization, and Model selections.
- ✓ Compare different paradigms for learning (Supervised, Unsupervised, etc.). Design experiments to evaluate and compare different machine learning techniques on real world problems.
- ✓ Employ probability, statistics, calculus, linear algebra, and optimization in order to develop new predictive models or learning method.

Course Fee: Rs: 2000/-

Eligibility

- 2nd Year Polytechnic Diploma in Computer Science/ IT/ Electronics /Electrical/ Instrumentation/
- or
- Final year BCA/B.Sc.(Electronics/CS/IT)
- or
- Pursuing MCA/MSC(CS/IT) /PGDCA
- or
- Pursuing B.E/B.Tech in any domain
- or
- NIELIT O Level (IT)

Methodology:

- ✓ Teaching Mode: Online
- ✓ Instructor-led live sessions
- ✓ Online/ Blended lab sessions
- ✓ Content access through LMS
- ✓ Recorded Session Available

Registration Link: <http://nva.nielit.gov.in>

Contact Details:

- Course Incharge : Dharendra kumar , STA(Mob- 7739757168)
- Course Coordinator Name: Mr. Dhanush
- Mobile number: 6383147051
- Email: trng.chennai@nielit.gov.in, dharendra@nielit.gov.in

Course Structure:

Basic and Python Ecosystem(4 Hours)

- Python Overview –Data Types, data structures, if statements, looping and loop control statements, Functions and Modules.

Data Loading for ML Projects (6 Hours)

- Introduction to Pandas library, built in functions, `read_csv`, `head()`, `describe()`, `memory_usage()`, `astype()`, `loc[:,]`, `value_counts()`, `drop_duplicates()`, `groupby()`, `fillna()` , `dropna()`, etc.
- Introduction to Numpy library, numpy built in functions, `add()`, `multiply()`, `divide()`, `floor()`, `mod()`, `sin()`, `radians()`, `log()`, `ceil()`, `sqrt()`, etc.

Understanding Data with Statistics (6 Hours)

- Descriptive And Inferential statistics
- Measures of Central Tendencies
- Mean, Mode, Median, Standard Deviation
- Z-Score, Outliers, Normal Distribution
- Correlation, Covariance and Regression Theory.
- Semantic HTML
- Linking CSS and JavaScript
- HTML Entities

Understanding Data with Visualization (6 Hours)

- Introduction to Matplotlib, built in functions to plot graph, bar, barh, figure, plot, pie, subplot, etc.
- Introduction to seaborn, visualize using different plots using seaborn built in functions Barplot, Countplot, distplot, Heatmap, Scatterplot, Pairplot, Boxplot.

Preparing Data (6 Hours)

- Scaling
- Normalization
- Handling Missing Data
- Encoding Categorical Data
- Handling Outliers.

Data Feature Selection(6 Hours)

- Introduction to Features Engineering.
- Feature Selection methods.
- Backward elimination technique .
- Forward selection technique.
- Feature importance from ML modeling.

Machine Learning Algorithms (30 Hours)

- Supervised Machine Learning Algorithms: Linear Regression, Logistic Regression, K Nearest Neighbor, Decision Tree, Naïve Bayes, Bagging, Boosting, SVM .
- Unsupervised Machine Learning Algorithms: K- Means Clustering, Principle

Component Analysis (PCA).

Machine Learning with Pipelines (4 Hours)

- Learn to improve accuracy by building different models.

Improving Performance of ML Models (6 Hours)

- K-Fold Cross Validation .
- Grid And Randomized Search Cv In Sklearn .
- Imbalanced Data Set : Smote Technique.

Evaluating and Interpreting ML Models (6 Hours)

- Introduction to Model Evaluation.
- Metrics for Classification Models .
- Metrics for Regression Models.
- Cross-Validation Techniques.
- Interpreting Model Results.

Examination & Certification

Paper Name	Certified Web Developer				Total
Category	Theory	Internal Assignments	Project / Presentation / Assignment	Practical	
Marks	100	20	20	60	200

Marks Range	Grade	Certificate Type
$\geq 85\%$	S	Graded
$\geq 75\% \text{ \& } < 85\%$	A	Graded
$\geq 65\% \text{ \& } < 75\%$	B	Graded
$\geq 55\% \text{ \& } < 65\%$	C	Graded
$\geq 50\% \text{ \& } < 55\%$	D	Graded
$< 50\%$	F	Participation
Attended the Course but not appeared in Examination	N	Participation